

v1.0Graviton Deep Dive Technical Report

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Graviton Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Graviton Deep Dive Technical Report V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17+} particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Graviton mass bound	Pre-O4 LIGO	LIGO O4 2024: $m_g < 1.27 \times 10^{-23}$ eV from GW dispersion. Best bound. Propagator updated	MEDIUM	Core graviton mass constraint updated from latest data
GW polarization modes	GR (+,×) only assumed	LIGO O4 2024: tensor modes dominant, vector/scalar $< 5\%$ (95% CL). GR confirmed. Non-GR modes $< 5\%$	HIGH	Polarization test directly validates graviton spin-2 assumption in simulation
Quantum gravity decoherence	Speculative	Pikovski 2024: gravitational decoherence testable with ET. V2.0: quantum noise term added near BH	LOW	Future-facing addition; not yet observable

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Massive gravity Vainshtein	Massless graviton assumed	V2.0: GR regime $r < r_V$, modified gravity $r > r_V$. Two spatial regimes now implemented	MEDIUM	Massive gravity theories require spatial mode switching not in V1.0

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** GW
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Graviton Deep Dive Technical Report V1.0 archived | V2.0 is the active specification as of June 2026